Zhengliang Shi

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EDUCATION

Shandong University | Bachelor of Computer Science and Technology *Sep 2019 - Jun 2023*

- GPA: 93.46 / 100, Rank: 2 / 182
- Core Modules: Machine Learning (99), Operating Systems (98), Advanced Mathematics (98), Advanced Design and Analysis of Algorithms (98), Discrete Mathematics (100)

RESEARCH INTERESTS

Natural Language Processing (Large language models, Knowledge discovery)

PUBLICATIONS (* equal contribution)

[1] Confucius: Iterative Tool Learning from Introspection Feedback by Easy-to-Difficult Curriculum. Shen Gao*, *Zhengliang Shi**, Minghang Zhu, Bowen Fang, Xin Xin, Pengjie Ren, Zhumin Chen, Jun Ma. ***Under review***

[2] Towards a Unified Framework for Scientific Reference Retrieval and Related Work Generation. *Zhengliang Shi,* Shen Gao, Xiuying Chen, Zhen Zhang, Pengjie Ren, Zhumin Chen, Zhaochun Ren. ***EMNLP*** 2023

[3] Reference-Assisted Dialogue Evaluation for Open-Domain Dialogue. *Zhengliang Shi*, Weiwei Sun, Shuo Zhang, Pengjie Ren, Zhumin Chen, Zhaochun Ren. Accepted by ***ACL*** 2023

[4] Contrastive Learning Reduces Hallucination in Conversations. Weiwei Sun, *Zhengliang Shi,* Shen Gao, Pengjie Ren, Zhumin Chen, Zhaochun Ren. Accepted by ***AAAI*** 2023

RESEARCH EXPERIENCE

Iterative Tool Learning with Foundation Models *Mar 2023 - Aug 2023*

Research Assistant | IRlab, Shandong University *Advisor: Prof. Zhaochun Ren (Leiden Uni.)*

- Propose the *Confucius*, a tool-learning framework, that enhances the large language model (LLM) with external tools to access diverse knowledge sources, includeing two main phases:
  (1) The multi-stage learning, teaching the LLM to use various tools from an easy-to-difficult curriculum
  (2) The **I**terative **S**elf-instruct from **I**ntrospective **F**eedback (ISIF), dynamically constructing the dataset to improve the ability to use the complicated tool
- Conduct extensive experiments on both controlled and real-world settings to demonstrate the superiority of our tool learning framework
- Release the tool-learning dataset to facilitate further research

Towards a More Reliable Evaluation for Open-Domain Dialogue *Jul 2022 - May 2023*

Research Assistant | IRlab, Shandong University *Advisor: Prof. Zhaochun Ren (Leiden Uni.)*

- Cooperate with Dr. Zhuo Shang (researcher in *Bloomberg*) and introduce the **R**eference-**A**ssisted **D**ialogue **E**valuation (RADE) method to build a trustworthy evaluation method for the open-domain dialogue system
- Extend three existing dialogue datasets with additional rated responses other than just a golden response by human annotation to support RADE

- Verify the effectiveness of our method on five benchmarks, where the Spearman, Pearson, and Kendall correlations with human evaluation outperform state-of-the-art baselines

Reduce the Hallucination of Language Models in Dialogue System

Research Assistant | IRLab, Shandong University. *Sep 2021 - Aug 2022*

Advisor: Prof. Zhaochun Ren (Leiden University), Prof. Maarten de Rijke (University of Amsterdam)

- Conduct the pilot experiment and group the hallucination of the dialogue system into two types, i.e., intrinsic and extrinsic hallucination
- Propose the mixed contrastive learning method to reduce the hallucination, including two steps:
  (1) Sample the most confused negative knowledge by retrieving from the corpus or deriving via model bootstrapping
  (2) Construct the negative sample via two mixed strategies, i.e., entity-based mix-up and constituency-based mix-up, regarding different types of hallucination
- Demonstrated that our proposed method effectively reduces the hallucinating content generated by the language model and improve the truthfulness of dialogue systems.

Other Projects

[1] Towards a Unified Framework for Reference Retrieval and Related Work Generation
Research Assistant | IRlab, Shandong University *Feb 2023 – Aug 2023*

[2] Learn to ask: Large language model is an Interactive Agent
Researcher | Intelligent Search and Strategy Group, Baidu *Sep 2023 - Present*

[3] Rethink before retrieval: Trustworthy Retrieval-argument Generation for Question Answering
Research Assistant | IRlab, Shandong University *Sep 2023 - Present*

LEADERSHIP EXPERIMENT

Teaching Assistance

Teaching Assistance Program, Shandong University *Sep 2022 - Present*

- Served as a Teaching Assistant for multiple computer science courses, which include (1) Operating Systems, (2) Databases, (3) Natural Language Processing

Team Leader

Research Internship Team, Shandong University *Sep 2022 - Present*

- Planed and facilitated weekly meetings and discussions
- Coordinated the assignments for the interns
- Organized and leading outreach events

Team Leader

Education Support Society (ESS), Shandong University. *Sep 2021 - Seq 2022*

- Led a team of *ESS* to facilitate both online and offline teaching projects
- Honored with the National Outstanding Team recognition

HONOR & AWARD (SELECTED)

National Scholarship, Ministry of Education, China *2021*

Dean's Scholarship, Shandong University, Department of Computer Science *2022*

Presidential Scholarship, Shandong University, China *2022*

Excellent Graduate, Shandong University, China *2023*

Second Prize of the American Mathematical Modeling Competition *2021, 2022*

Third Prize of World Artificial Intelligence Challenge Competition (WAIC) *2022*